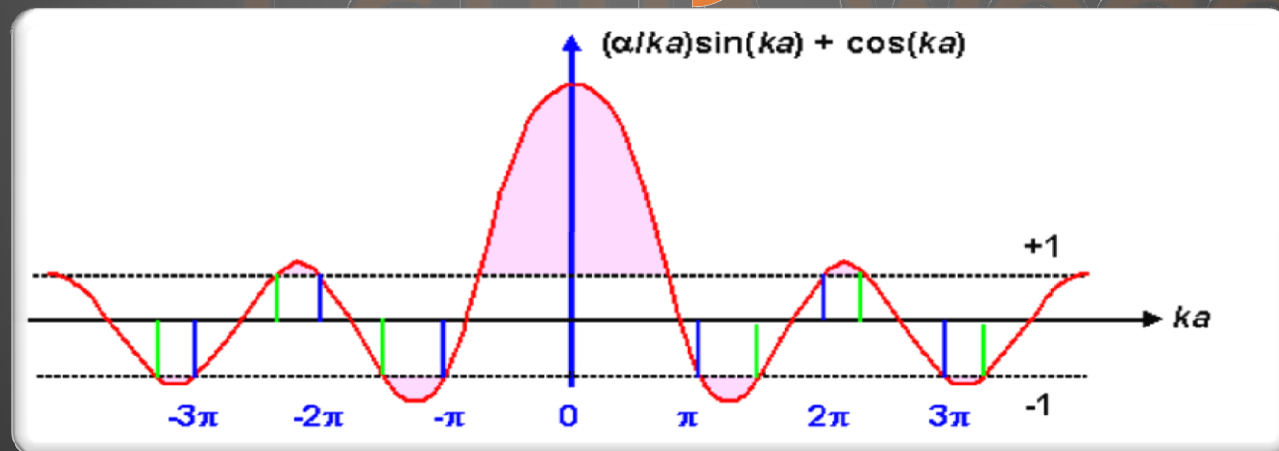


The Kronig Penny Model

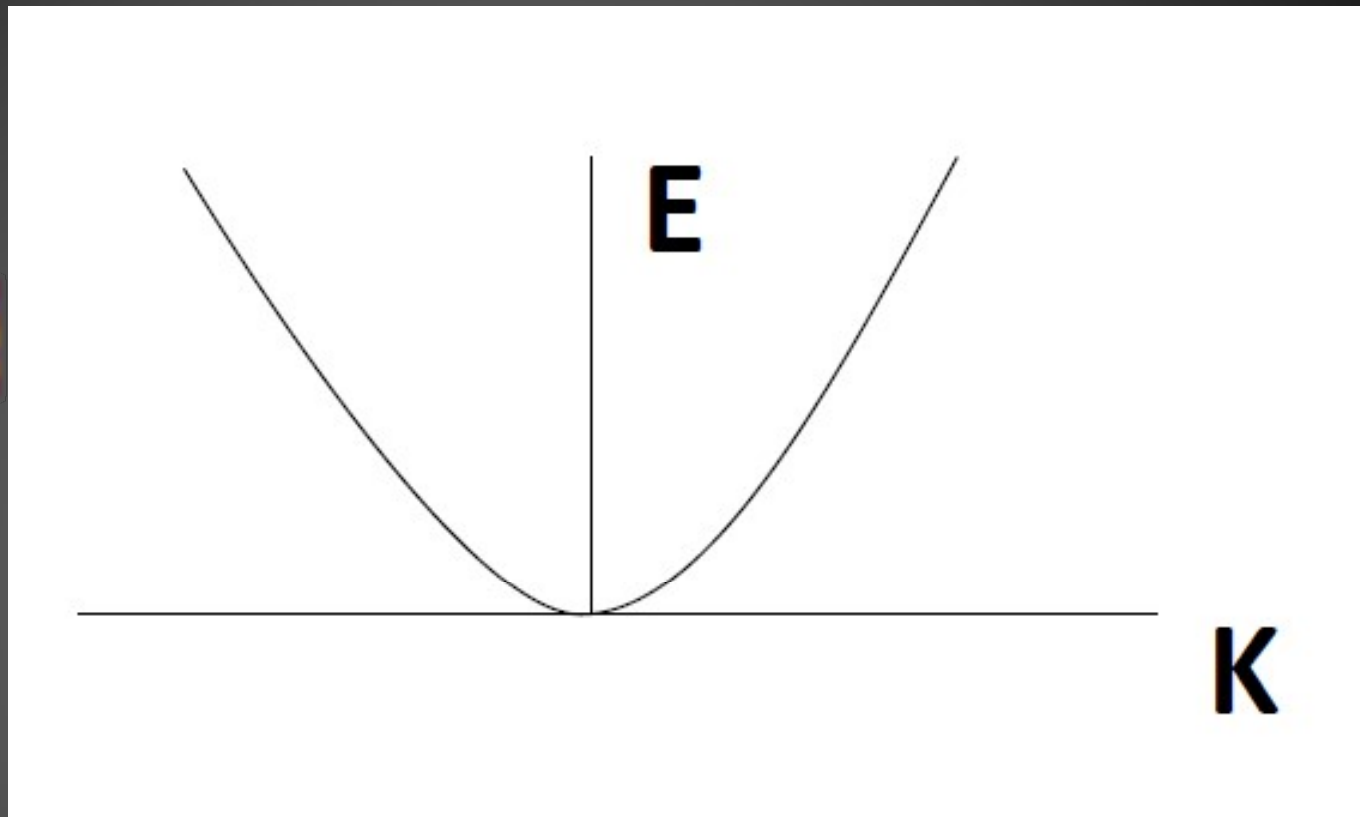


Nafees Uddin
Assistant Professor
Applied Physics

Limitations of Free Electron Theory

According to quantum free electron theory of metals, a conduction electron in a metal experiences constant (or zero) potential and free to move inside the crystal but will not come out of the metal because an infinite potential exists at the surface.





EK Curve for Free Electron





This theory successfully explains :

- ✓ electrical conductivity
- ✓ specific heat,
- ✓ thermionic emission
- ✓ paramagnetism.





Failures



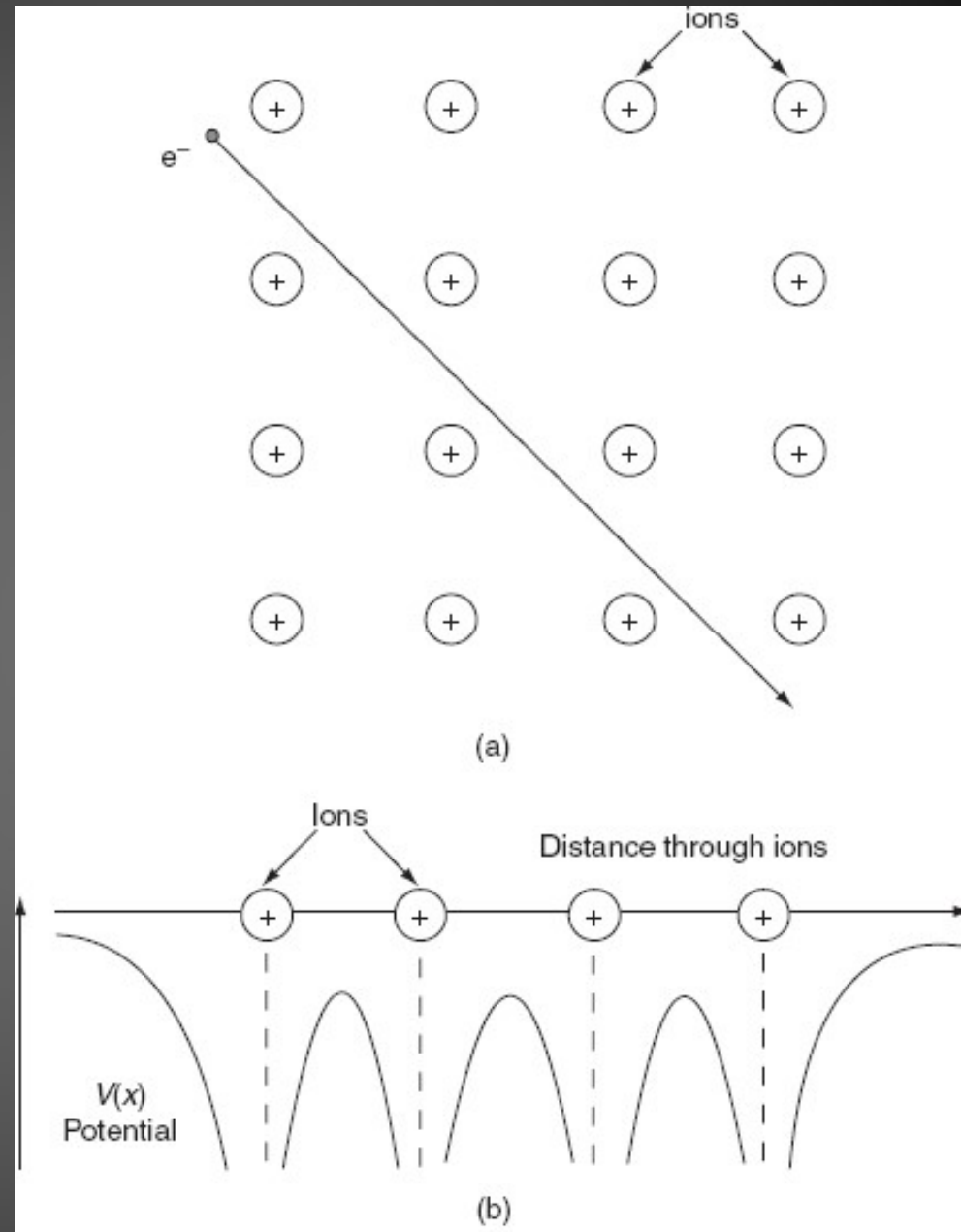
This theory fails to explain many other physical properties, for example:

- (i) it fails to explain the difference between conductors, insulators and semiconductors,
- (ii) positive Hall coefficient of metals and
- (iii) lower conductivity of divalent metals than monovalent metals.



Reason

Electrons face a periodic potential while moving through crystal lattice.



Bloch Theorem

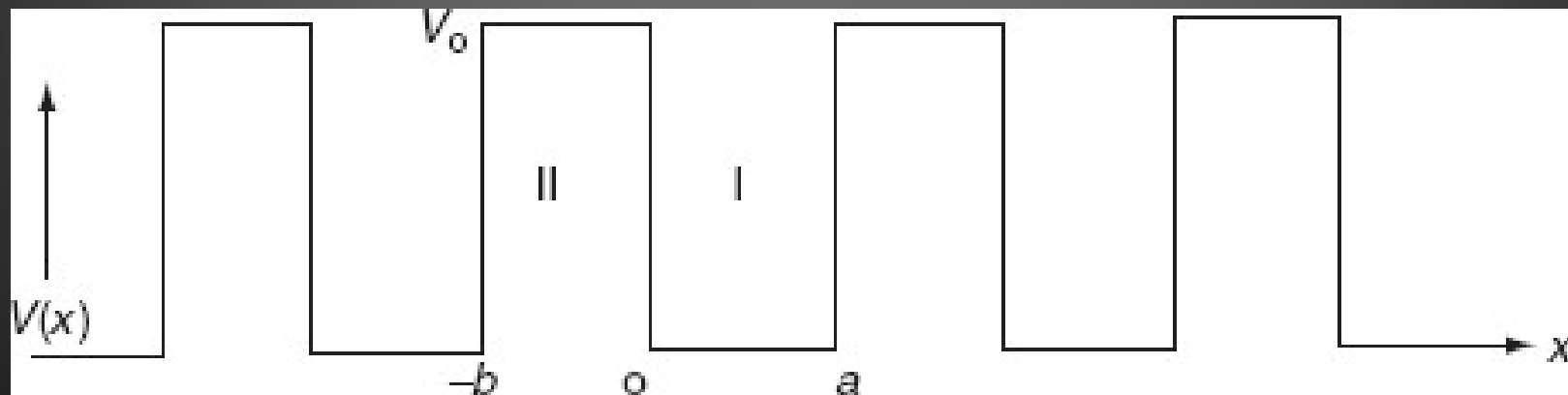
- Asserts that solution in periodic potential is always a product of two terms:
 - a periodic function (with the same period)
 - a plane wave
- Mathematically, we can write:

$$\Psi(\mathbf{r}) = Ae^{i\mathbf{k}\cdot\mathbf{r}}u(\mathbf{r})$$

where $u(\mathbf{r} + \mathbf{R}) = u(\mathbf{r})$

Kronig Penney Model

- It is not easy to solve Schrödinger's equation with these potentials. So, Kronig and Penney approximated these potentials inside the crystal to the shape of rectangular steps as shown in Fig. This model is called Kronig-Penney model of potentials.



Kronig Penney Model

- The potential energy is zero in the region $0 < x < a$ and V_0
 - In the region $-b < x < 0$
 - $V(x) = 0 \quad : 0 < x < a$
 - $V(x) = V_0 \quad : -b < x < 0$
- 1.
- The Schrodinger equation for the two regions become:

$$\frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} E \psi = 0 \quad \text{for } 0 < x < a \dots\dots\dots (1)$$

$$\frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} [E - V_0] \psi = 0 \quad \text{for } -b < x < 0$$

Kronig Penney Model

- We define two real quantities (say) α and β such that:

$$\alpha^2 = \frac{2mE}{\hbar^2} \quad \text{and} \quad \beta^2 = \frac{2m}{\hbar^2}(V_0 - E)$$



Kronig Penney Model

- Schrodinger equation now becomes:

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \quad \text{for } 0 < x < a$$
$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \quad \text{for } -b < x < 0$$

Kronig Penney Model

- Using the Bloch theorem

$$\psi(x) = e^{ikx} u(x).$$

The Schrodinger equations are solved to get:

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos Ka$$

where $P = \frac{mV_0ab}{\hbar^2}$

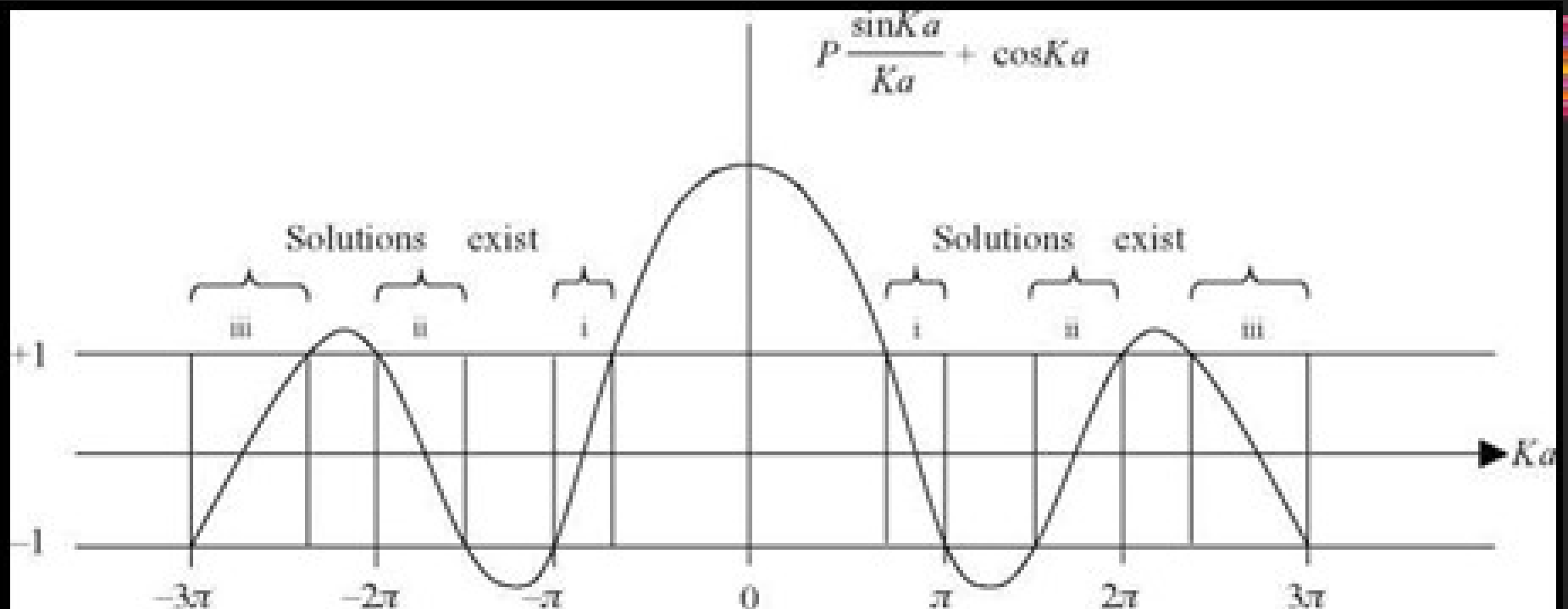
Kronig Penney Model

- P is a measure of barrier strength.
- Two cases arise:
- Case I:
- $P \rightarrow \infty \sin \alpha a = 0$ or $\alpha = n\pi/a$
- $\alpha^2 = (n\pi/a)^2$ or
- $E = n^2 h^2 / 8ma^2$
- Which is the expression for of particle in a box

Kronig Penney Model

- Case II:
- $P \rightarrow 0 \cos ka = ka$ or $\alpha = k$
- Or
- $E = \frac{k^2 h^2}{2m}$
- Which is the expression for energy of a free electron

Kronig Penney Model



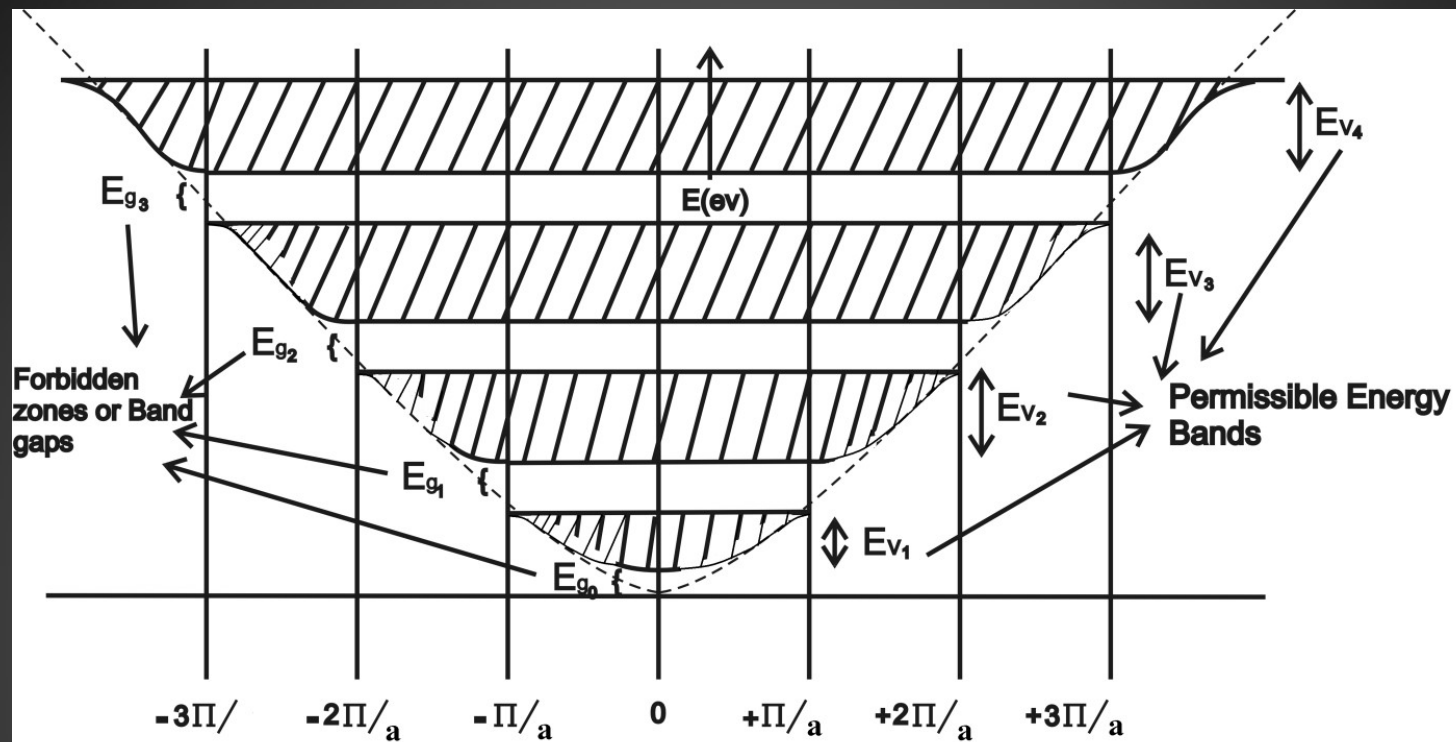
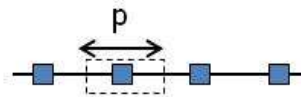


Fig 1.49 Energy Band Diagram of electrons in a crystal solid -the interruption of E-k diagram due to Bragg Reflection

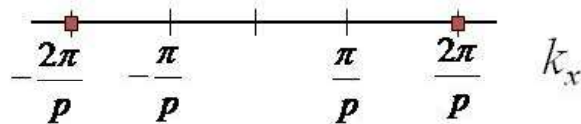
Brillouin Zone for One-dimensional Solids

Real-space



Replacing
(a+b) by p ...

1st B-Z



E-k diagram

